

Kunal Kaushik

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Work Experience

Vision Researcher, Prof. Armin Moghadam at SJSU

Jun 2024 to Nov 2025

- Developed **YOLO V9 model** for wafer alignment during semiconductor manufacturing; **95.0% overall accuracy**
 - Utilized Google Colab and **Python** for training, preparing for submission to **Springer Intelligent Manufacturing Journal**
- Formulated a computer vision pipeline to annotate **mitochondrial networks** in endothelial cells with Journey Lab
 - **ImageJ** for data processing, with **U-net and SegFormer** for semantic segmentation model training

Mechatronics Intern, Nova Robotics LLC

Jan 2025 to Jul 2025

- Designed 3D-printed **self-balancing humanoid** robot with gyroscopic **dual reaction wheel** system
 - Prototyped reaction wheel mechanism using **inverted pendulum** setup as proof of concept
- **Raspberry Pi** as microcontroller + vision processing, **MPU9050 IMU** for balance, + **ODrive S1** motor driver for 6 DoF
- Reduced weight to **<10kg** and cost to **\$6,000** to polish for both research + **child/elderly care** applications

Software Engineering Intern, AliveCor

Jun 2025 to Jul 2025

- Added **device interaction** (credit management, blocking devices) with **Go and Javascript** for internal customer service capabilities
- Streamlined **Kardia 6L** workflows to improve **verification process** for higher efficacy levels during prototyping
- Presented work to **50+ employees** and **executive board** of AliveCor in-person at conference

Software Engineering Intern, Marchup.net

Jan 2024 to May 2024

- Built automated **Selenium testing framework** to validate website and mobile app functionality across core user flows
- Debugged and optimized messaging and media feed systems using **React and JavaScript**, improving stability and performance
- Served as **school ambassador**, coordinating with **administrators** to integrate platform features into existing school workflows and systems

Projects

Founder and Captain, Rift Robotics #26024

Jun 2024 to Nov 2025

- Founded and led a **7-person FTC team**, raising **\$3,000+** with crowdfunding and AoPS sponsorship
- Designed + assembled **3000+ part drivetrain**; custom CNC milled and **3D-printed parts** for optimized model
- **Pedro Pathing algorithm** for tele-operated + autonomous periods enabled one-button cycling
- Earned Control Award, Finalist Alliance, and Inspire 2nd Award as **community team**

SnoSight AR Ski Goggles Kit

Jul 2024 to Jul 2025

- Architected a **retrofitting kit** that incorporates **augmented reality** + phone connectivity into existing ski goggles
- Used an **ESP32** to enable low-latency BLE/WiFi syncing between goggles and a **Flutter iOS app** with live GPS mapping
- Integrated an IMU for **real-time ski analytics** and built an **avalanche-detection** algorithm that auto-triggers emergency alerts

Education

Wilcox High School

Aug 2022 to Present

Diploma

GPA: 4.000 / 4.5789, Top 10%

Relevant Coursework: Calculus AB, Calculus BC, Computer Science P, Computer Science A, Physics 1, Physics 2

Dual Enrollment: Calculus 3, Robotics and Embedded Systems, Database Management Systems I

Activities: Computer Science Club, Artificial Intelligence Club, Varsity Tennis

Technical Proficiencies

EDA/CAD:	Fusion 360, Onshape, EasyEDA, KiCad
Programming:	Python, C/C++, Java, Go, Flutter, SQL
Embedded:	ESP32, Raspberry Pi, Arduino, STM32
Manufacturing:	CNC, 3D-printing, Soldering, Crimping, DFM, DFA